

Lesson 3

Part I: Vectors

Definition: An ordered list of numbers is called a **vector**.

In linear algebra, if a matrix has only one row or only one column it is called a vector. For example, let u_1 and u_2 be real numbers.

- $\mathbf{u} = \begin{bmatrix} u_1 \\ u_2 \end{bmatrix}$: a column vector in $\mathbb{R}^2$
- $\mathbf{u} = [u_1 \ u_2]$: a row vector

Operations on Vectors

Let $\mathbf{u} = \begin{bmatrix} u_1 \\ u_2 \end{bmatrix}$ and $\mathbf{v} = \begin{bmatrix} v_1 \\ v_2 \end{bmatrix}$ be two vectors in $\mathbb{R}^2$.

1. $\mathbf{u} = \mathbf{v} \iff$
2. $\mathbf{u} + \mathbf{v} = \begin{bmatrix} u_1 \\ u_2 \end{bmatrix} + \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} =$
3. $c\mathbf{u} = c \begin{bmatrix} u_1 \\ u_2 \end{bmatrix} =$

For example, if $\mathbf{u} = \begin{bmatrix} 1 \\ 3 \end{bmatrix}$ and $\mathbf{v} = \begin{bmatrix} 2 \\ -1 \end{bmatrix}$, then

$$3\mathbf{u} - \mathbf{v} =$$

Vectors in $\mathbb{R}^n$ can be written as $\mathbf{u} = \begin{bmatrix} u_1 \\ u_2 \\ \vdots \\ u_n \end{bmatrix}$.

The **zero vector** (denoted by $\mathbf{0}$) in $\mathbb{R}^n$ is the vector whose n entries are all zero.

Properties of Vector Operations

For vectors $\mathbf{u}, \mathbf{v}, \mathbf{w}$ in $\mathbb{R}^n$ and for scalars c and d :

1. $\mathbf{u} + \mathbf{v} = \mathbf{v} + \mathbf{u}$
2. $(\mathbf{u} + \mathbf{v}) + \mathbf{w} = \mathbf{u} + (\mathbf{v} + \mathbf{w})$
3. $\mathbf{u} + \mathbf{0} = \mathbf{0} + \mathbf{u} = \mathbf{u}$
4. $\mathbf{u} + (-\mathbf{u}) = -\mathbf{u} + \mathbf{u} = \mathbf{0}$, where $-\mathbf{u} = (-1)\mathbf{u}$
5. $c(\mathbf{u} + \mathbf{v}) = c\mathbf{u} + c\mathbf{v}$
6. $(c + d)\mathbf{u} = c\mathbf{u} + d\mathbf{u}$
7. $c(d\mathbf{u}) = (cd)\mathbf{u}$
8. $1\mathbf{u} = \mathbf{u}$

Geometric Description of vectors in $\mathbb{R}^2$

1. The column vector $\mathbf{u} = \begin{bmatrix} u_1 \\ u_2 \end{bmatrix}$ can be graphed as a directed line segment from the origin $(0, 0)$ to the point (u_1, u_2) . It is called the position vector.

2. The sum of two vectors $\mathbf{u} + \mathbf{v}$: use the Parallelogram Rule

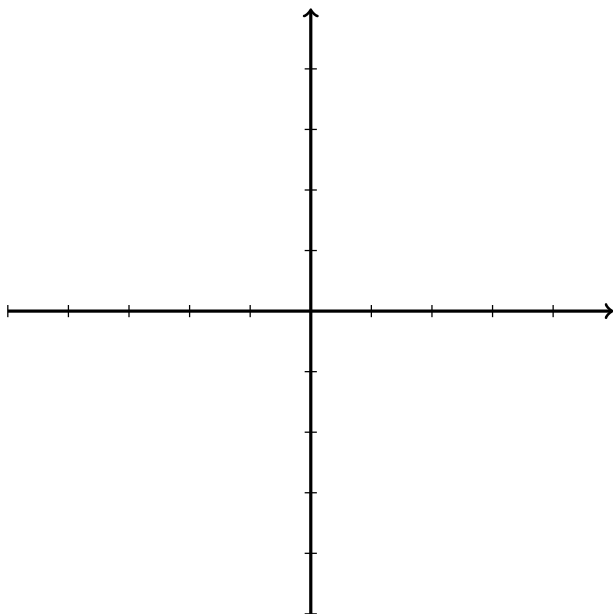
3. Scalar multiplication of a vector $c\mathbf{u}$:

(i) $c > 0$:

(ii) $c < 0$:

Note: The set of all scalar multiples of a nonzero vector $\mathbf{u}$ is a line through the origin and $\mathbf{u}$.

Example 1: Given two vectors $\mathbf{u} = \begin{bmatrix} 1 \\ 3 \end{bmatrix}$ and $\mathbf{v} = \begin{bmatrix} 2 \\ -1 \end{bmatrix}$, find the vectors $\mathbf{u} + \mathbf{v}$, $2\mathbf{v}$, $-\mathbf{v}$ and display them on a graph.



Part II: Vector Equations

Linear Combinations

Definition: Given vectors $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_p \in \mathbb{R}^n$ and scalars $c_1, c_2, \dots, c_p$, the vector $\mathbf{b}$ defined by

$$\mathbf{b} = c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + \dots + c_p\mathbf{v}_p$$

is called a **linear combination** of $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_p$ with weights $c_1, c_2, \dots, c_p$.

For Example, let $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_p \in \mathbb{R}^n$.

(1) A linear combination of $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_p$ with weights $c_1 = c_2 = \dots = c_p = 0$ is

(2) A linear combination of $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_p$ with weights $c_1 = 1, c_2 = \dots = c_p = 0$ is

(3) A linear combination of $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_p$ with $c_i = 1$ and other weights $c_j = 0$ (if $j \neq i$) is

Statement: Let $\mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_n$, and $\mathbf{b}$ be vectors in $\mathbb{R}^m$. $\mathbf{b}$ can be written as a linear combination of $\mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_n$ if and only if there exist weights $x_1, x_2, \dots, x_n$ such that

$$x_1\mathbf{a}_1 + x_2\mathbf{a}_2 + \dots + x_n\mathbf{a}_n = \mathbf{b} \tag{1}$$

that is, the **vector equation** (1) has a solution.

Statement: A vector equation

$$x_1 \mathbf{a}_1 + x_2 \mathbf{a}_2 + \cdots + x_n \mathbf{a}_n = \mathbf{b}$$

has the same solution set as the linear system $A\mathbf{x} = \mathbf{b}$.

Moreover, $\mathbf{b}$ can be generated by a linear combination of columns of A if and only if there exists a solution to the linear system $A\mathbf{x} = \mathbf{b}$.

Definition: If $\mathbf{v}_1, \mathbf{v}_2, \cdots, \mathbf{v}_p \in \mathbb{R}^n$, then

$$\text{Span} \{ \mathbf{v}_1, \cdots, \mathbf{v}_p \} = \{ c_1 \mathbf{v}_1 + \cdots + c_p \mathbf{v}_p \mid c_1, \cdots, c_p \text{ are scalars.} \}$$

is the subset of $\mathbb{R}^n$ **spanned** by $\mathbf{v}_1, \mathbf{v}_2, \cdots, \mathbf{v}_p$.

For example, let $S = \text{Span} \{ \mathbf{v}_1, \cdots, \mathbf{v}_p \} = \{ c_1 \mathbf{v}_1 + \cdots + c_p \mathbf{v}_p \mid c_i \in \mathbb{R} \}$.

Then (1) $\mathbf{0} \in S$

$$(2) \mathbf{v}_1 \in S$$

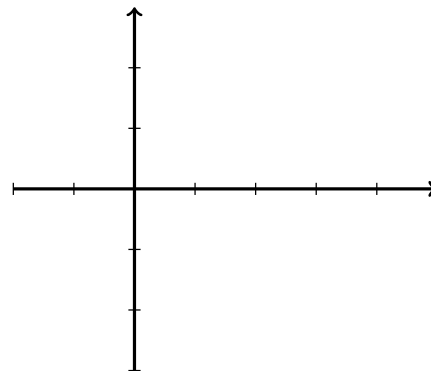
$$(3) c\mathbf{v}_1 \in S$$

Conclusion: S contains every scalar multiple of a vector $\mathbf{v}_i$ ($i = 1 : p$).

Statement: The following statements are logically equivalent:

1. A vector $\mathbf{b}$ is in $\text{Span}\{\mathbf{a}_1, \dots, \mathbf{a}_n\}$.
2. A vector equation $x_1\mathbf{a}_1 + x_2\mathbf{a}_2 + \dots + x_n\mathbf{a}_n = \mathbf{b}$ has a solution.
3. A linear system $A\mathbf{x} = \mathbf{b}$ with augmented matrix $[\mathbf{a}_1 \ \mathbf{a}_2 \ \dots \ \mathbf{a}_n \ \mathbf{b}]$ is consistent.

Example 2: Let $\mathbf{a}_1 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$, $\mathbf{a}_2 = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$, and $\mathbf{b} = \begin{bmatrix} 4 \\ -1 \end{bmatrix}$. Determine whether $\mathbf{b}$ can be written as a linear combination of $\mathbf{a}_1$ and $\mathbf{a}_2$. If yes, find weights and display the vectors on a graph.



Geometric Description of a Spanned Set in $\mathbb{R}^3$

1. $\text{Span}\{\mathbf{u}\}$

(i) If $\mathbf{u} = \mathbf{0}$, then $\text{Span}\{\mathbf{u}\} =$

(ii) If $\mathbf{u} \neq \mathbf{0}$, then $\text{Span}\{\mathbf{u}\} =$

2. $\text{Span}\{\mathbf{u}, \mathbf{v}\}$

Assume that $\mathbf{u}, \mathbf{v} \neq \mathbf{0}$.

(i) If $\mathbf{v} = c\mathbf{u}$, then $\text{Span}\{\mathbf{u}, \mathbf{v}\} =$

(ii) If $\mathbf{v} \neq c\mathbf{u}$, then $\text{Span}\{\mathbf{u}, \mathbf{v}\} =$

3. $\text{Span}\{\mathbf{u}, \mathbf{v}, \mathbf{w}\} =$

Example 3: Let $\mathbf{a}_1 = \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix}$, $\mathbf{a}_2 = \begin{bmatrix} 2 \\ 1 \\ 3 \end{bmatrix}$, and $\mathbf{b} = \begin{bmatrix} 4 \\ -1 \\ 2 \end{bmatrix}$.

(1) What is $\text{Span}\{\mathbf{a}_1, \mathbf{a}_2\}$?

(2) Is $\mathbf{b}$ in $\text{Span}\{\mathbf{a}_1, \mathbf{a}_2\}$?

Lesson 4

Part I: The Matrix-Vector Product

Definition: Let $A = [\mathbf{a}_1 \ \mathbf{a}_2 \ \cdots \ \mathbf{a}_n]$ be an $m \times n$ matrix and $\mathbf{x} \in \mathbb{R}^n$. The **product of A and $\mathbf{x}$** is the linear combination of the columns of A using the corresponding entries in $\mathbf{x}$ as weights.

$$A\mathbf{x} = [\mathbf{a}_1 \ \mathbf{a}_2 \ \cdots \ \mathbf{a}_n] \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix} \stackrel{\text{def}}{=} x_1\mathbf{a}_1 + x_2\mathbf{a}_2 + \cdots + x_n\mathbf{a}_n$$

Note: The product $A\mathbf{x}$ is defined only when the number of columns in A is equal to the number of entries in $\mathbf{x}$.

Definition: An $n \times n$ **identity matrix** is defined as

$$I = \begin{bmatrix} 1 & 0 & 0 & \cdots & 0 \\ 0 & 1 & 0 & \cdots & 0 \\ \vdots & & & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & 1 \end{bmatrix} = [\mathbf{e}_1 \ \mathbf{e}_2 \ \cdots \ \mathbf{e}_n].$$

For example, consider a product of a 3×3 identity matrix I and

$$\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} :$$

Note: If I is an $n \times n$ identity matrix, $I\mathbf{x} = \mathbf{x}$ for any $\mathbf{x} \in \mathbb{R}^n$.

Properties of the Matrix-Vector Product

Theorem: If A is an $m \times n$ matrix, $\mathbf{u}$ and $\mathbf{v}$ are vectors in $\mathbb{R}^n$, and c is a scalar, then

1. $A(\mathbf{u} + \mathbf{v}) = A\mathbf{u} + A\mathbf{v}$
2. $A(c\mathbf{u}) = c(A\mathbf{u})$

How to Calculate the Matrix-Vector Product?

1. Use the definition:

Example 1: If $A = \begin{bmatrix} 1 & 2 & 3 \\ -1 & 3 & 4 \\ 2 & -3 & -5 \end{bmatrix}$ and $\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$, calculate

$$A\mathbf{x} \stackrel{\text{def}}{=}$$

Note: If $\mathbf{x} = [x_i]$ and $\mathbf{y} = [y_i]$ are vectors in $\mathbb{R}^n$, the **dot product** of $\mathbf{x}$ and $\mathbf{y}$ is

$$\mathbf{x} \cdot \mathbf{y} \stackrel{\text{def}}{=} x_1y_1 + x_2y_2 + \cdots + x_ny_n.$$

2. **Row-Vector Rule** for $A\mathbf{x}$:

When the product $A\mathbf{x}$ is defined, the i th entry in $A\mathbf{x}$ is the dot product of the i th row of A and the vector $\mathbf{x}$.

Example 2: Calculate the product:

$$(1) \begin{bmatrix} 2 & 0 & 6 \\ 3 & 1 & 4 \end{bmatrix} \begin{bmatrix} 1 \\ 5 \end{bmatrix} =$$

$$(2) \begin{bmatrix} 1 & 2 & -2 \\ 0 & 3 & 5 \end{bmatrix} \begin{bmatrix} 1 \\ -2 \\ 1 \end{bmatrix} =$$

$$(3) \begin{bmatrix} 3 & 1 \\ 2 & 4 \\ -1 & 0 \end{bmatrix} \begin{bmatrix} 2 \\ 5 \end{bmatrix} =$$

Part II: Matrix Equations

Definition: An equation of the form $A\mathbf{x} = \mathbf{b}$ is called a **matrix equation**.

We can write a linear system in three ways:

(1) as linear equations

$$\begin{array}{rcccccl} 2x_1 & + & 3x_2 & + & 4x_3 & = & -6 \\ x_1 & & & - & 2x_3 & = & 9 \\ & & 2x_2 & - & 3x_3 & = & 0 \end{array}$$

(2) as a vector equation

(3) as a matrix equation

Theorem: (Test for Consistency)

Let A be an $m \times n$ matrix. The following are logically equivalent:

1. The matrix equation $A\mathbf{x} = \mathbf{b}$ is consistent.
2. The vector equation $x_1\mathbf{a}_1 + x_2\mathbf{a}_2 + \cdots + x_n\mathbf{a}_n = \mathbf{b}$ has a solution.
3. $\mathbf{b}$ is in $\text{Span}\{\mathbf{a}_1, \mathbf{a}_2, \cdots, \mathbf{a}_n\}$.
4. The linear system with augmented matrix $[\mathbf{a}_1 \ \mathbf{a}_2 \ \cdots \ \mathbf{a}_n \ \mathbf{b}]$ is consistent.

Definition: Let each $\mathbf{v}_i \in \mathbb{R}^m$. We say that a set $\{\mathbf{v}_1, \mathbf{v}_2, \cdots, \mathbf{v}_p\}$ **spans** $\mathbb{R}^m$ if every vector in $\mathbb{R}^m$ is a linear combination of $\mathbf{v}_1, \mathbf{v}_2, \cdots, \mathbf{v}_p$, and we write

$$\mathbb{R}^m = \text{Span}\{\mathbf{v}_1, \cdots, \mathbf{v}_p\}.$$

For example, if $\mathbf{v}_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$, $\mathbf{v}_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$, then $\mathbb{R}^2 = \text{Span}\{\mathbf{v}_1, \mathbf{v}_2\}$.

Theorem: Let A be an $m \times n$ matrix. The following statements are logically equivalent:

1. For each $\mathbf{b} \in \mathbb{R}^m$, the equation $A\mathbf{x} = \mathbf{b}$ has a solution.
2. Each $\mathbf{b}$ in $\mathbb{R}^m$ is a linear combination of the columns of A .
3. The columns of A span $\mathbb{R}^m$.
4. A has a pivot position in every row.
5. $\text{rank}(A) = m$.

Example 3: Let $A = \begin{bmatrix} 1 & 2 & 3 \\ -1 & 3 & 4 \\ 2 & -6 & -7 \end{bmatrix}$ and $\mathbf{b} = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix}$.

- (1) Is the equation $A\mathbf{x} = \mathbf{b}$ consistent for all possible b_1, b_2, b_3 ?
- (2) Do the columns of A span $\mathbb{R}^3$?

Example 4: Let $A = \begin{bmatrix} 1 & 2 & 3 \\ -1 & 3 & 4 \\ 2 & -6 & -8 \end{bmatrix}$ and $\mathbf{b} = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix}$.

- (1) Is the equation $A\mathbf{x} = \mathbf{b}$ consistent for all possible b_1, b_2, b_3 ?
- (2) Do the columns of A span $\mathbb{R}^3$?

Lesson 5

Part I: Homogeneous Linear Systems

Definition: A linear system is called **homogeneous** if it can be written in the form $A\mathbf{x} = \mathbf{0}$, where A is an $m \times n$ matrix and $\mathbf{0} \in \mathbb{R}^m$ is the zero vector.

Note: A homogeneous linear system $A\mathbf{x} = \mathbf{0}$ always has at least one solution, $\mathbf{x} = \mathbf{0}$ (the zero vector in $\mathbb{R}^n$). It is called the **trivial solution**. Therefore, a homogeneous system is consistent.

Statement: A homogeneous equation $A\mathbf{x} = \mathbf{0}$ has a **nontrivial solution** if and only if the equation has at least one free variable.

For example, consider a homogeneous equation $A\mathbf{x} = \mathbf{0}$, where

$$A = \begin{bmatrix} 1 & -1 \\ -2 & 2 \end{bmatrix}.$$

Summary: For a homogeneous equation $A\mathbf{x} = \mathbf{0}$:

1. If the system has no free variable, it has only the trivial solution and the solution set is $\text{Span}\{\mathbf{0}\}$.
2. If the system has one free variable, its solution can be expressed as a parametric vector equation

$$\mathbf{x} = t\mathbf{v}, \quad t \in \mathbb{R}$$

and the solution set is $\text{Span}\{\mathbf{v}\}$.

3. If the system has two free variables, its solution can be expressed as a parametric vector equation

$$\mathbf{x} = s\mathbf{u} + t\mathbf{v}, \quad s, t \in \mathbb{R}$$

and the solution set is $\text{Span}\{\mathbf{u}, \mathbf{v}\}$.

Example 1: The augmented matrix of $A\mathbf{x} = \mathbf{0}$ has been row reduced to the RREF below. Find the solution set in parametric vector form and describe it geometrically.

$$\begin{bmatrix} 1 & 0 & 2 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

Example 2: Describe all solutions of the homogeneous system:

$$x_1 - 3x_2 + 2x_3 = 0$$

Part II: Nonhomogeneous Linear Systems

Definition: A linear system is called **nonhomogeneous** if it can be written in the form $A\mathbf{x} = \mathbf{b}$, where $\mathbf{b} \neq \mathbf{0}$.

Compare solutions for homogeneous and nonhomogeneous systems:

Theorem: Let $\mathbf{v}_h$ be a general solution of the homogeneous equation $A\mathbf{x} = \mathbf{0}$. Suppose $A\mathbf{x} = \mathbf{b}$ is consistent for a given vector $\mathbf{b}$ and let $\mathbf{p}$ be a solution to $A\mathbf{x} = \mathbf{b}$. Then the solution set of $A\mathbf{x} = \mathbf{b}$ is the set of all vectors in the form $\mathbf{x} = \mathbf{p} + \mathbf{v}_h$.

Summary: Geometrically, if $A\mathbf{x} = \mathbf{b}$ has a solution $\mathbf{p}$,

1. and if $A\mathbf{x} = \mathbf{0}$ has no free variable, its solution $\mathbf{p}$ is unique and the solution set is a single point $\mathbf{p}$.

2. and if $A\mathbf{x} = \mathbf{0}$ has one free variable, its solution is

$$\mathbf{x} = \mathbf{p} + t\mathbf{v}, \quad t \in \mathbb{R}$$

and the solution set is a line through $\mathbf{p}$ parallel to the line spanned by $\mathbf{v}$.

3. and if $A\mathbf{x} = \mathbf{0}$ has two free variables, its solution is

$$\mathbf{x} = \mathbf{p} + s\mathbf{u} + t\mathbf{v}, \quad s, t \in \mathbb{R}$$

and the solution set is a plane through $\mathbf{p}$ parallel to the plane spanned by $\mathbf{u}$ and $\mathbf{v}$.

Example 3: Find the solutions of the system in parametric vector form and describe the solution set geometrically.

$$x_1 + 2x_2 - 3x_3 = 5$$

$$2x_1 + x_2 - 3x_3 = 13$$

$$-x_1 + x_2 = -8$$

Example 4: Find the solutions of the system in parametric vector form and describe the solution set geometrically.

$$2x_1 - 4x_2 + 6x_3 = 8$$

Lesson 6

Part I: Linear Independence

Definition:

- A set of vectors $\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_p\}$ in $\mathbb{R}^n$ is **linearly independent** if the vector equation

$$x_1\mathbf{v}_1 + x_2\mathbf{v}_2 + \dots + x_p\mathbf{v}_p = \mathbf{0} \quad (1)$$

has only the trivial solution.

- If there is a nontrivial solution to the vector equation (1), then the set $\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_p\}$ is **linearly dependent**.

Note: Write $A = [\mathbf{v}_1 \ \mathbf{v}_2 \ \dots \ \mathbf{v}_p]$.

1. The vector equation (1) can be written as $A\mathbf{x} = \mathbf{0}$, and therefore, the set $\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_p\}$ is linearly independent if $A\mathbf{x} = \mathbf{0}$ has only the trivial solution. ($\iff A\mathbf{x} = \mathbf{0}$ has)

2. The set $\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_p\}$ is linearly dependent $\iff A\mathbf{x} = \mathbf{0}$ has

3. When $\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_p\}$ is linearly dependent, the equation

$$c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + \dots + c_p\mathbf{v}_p = \mathbf{0}, \ c_1, \dots, c_p \text{ not all zero}$$

is a **linear dependence relation** among $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_p$.

Example 1: Let $\mathbf{v}_1 = \begin{bmatrix} 1 \\ -2 \\ 0 \end{bmatrix}$, $\mathbf{v}_2 = \begin{bmatrix} -2 \\ 4 \\ 1 \end{bmatrix}$, and $\mathbf{v}_3 = \begin{bmatrix} 3 \\ -6 \\ -1 \end{bmatrix}$.

(1) Show that $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$ is linearly dependent.

(2) Find a linear dependence relation among $\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3$.

Statement: The columns of matrix A are linearly independent
 $\iff A\mathbf{x} = \mathbf{0}$ has only the trivial solution.

Example 2: Determine if the columns of matrix A are linearly independent.

$$A = \begin{bmatrix} 1 & 2 & 3 \\ -2 & 4 & 5 \\ 3 & -2 & 1 \end{bmatrix}$$

Part II: Special Cases for Linear Dependence

1. A set of one vector: $\{\mathbf{v}_1\}$

$\{\mathbf{v}_1\}$ is linearly dependent $\iff \mathbf{v}_1 = \mathbf{0}$.

2. A set of two vectors: $\{\mathbf{v}_1, \mathbf{v}_2\}$

$\{\mathbf{v}_1, \mathbf{v}_2\}$ is linearly dependent $\iff \mathbf{v}_2 = c\mathbf{v}_1$.

3. A set of two or more vectors: $\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_p\}$

Theorem: $S = \{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_p\}$ is linearly dependent

$\iff$ at least one of the vectors in S is a linear combination of the others.

Special cases when linear dependence is automatic

Theorem:

If S contains more vectors than there are entries in each vector, then S is linearly dependent. That is, any set $S = \{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_p\}$ in $\mathbb{R}^n$ is linearly dependent if $p > n$.

Theorem:

If a set S in $\mathbb{R}^n$ contains the zero vector, then S is linearly dependent.

Geometric representation of dependence relations in $\mathbb{R}^3$

Example 3: Describe each set geometrically in $\mathbb{R}^3$.

(1) $\text{Span}\{\mathbf{u}, \mathbf{v}\}$, where $\mathbf{u} = \begin{bmatrix} 1 \\ 3 \\ 5 \end{bmatrix}$, $\mathbf{v} = \begin{bmatrix} -2 \\ -6 \\ -10 \end{bmatrix}$

(2) $\text{Span}\{\mathbf{u}, \mathbf{v}, \mathbf{w}\}$, $\mathbf{u} = \begin{bmatrix} 1 \\ 0 \\ -3 \end{bmatrix}$, $\mathbf{v} = \begin{bmatrix} 0 \\ -1 \\ 4 \end{bmatrix}$, $\mathbf{w} = \begin{bmatrix} 2 \\ -1 \\ -2 \end{bmatrix}$

Example 4: Determine by inspection if the given set is linearly dependent.

$$(1) \begin{bmatrix} 1 \\ 2 \\ 3 \\ 4 \end{bmatrix}, \begin{bmatrix} -2 \\ -4 \\ -6 \\ -9 \end{bmatrix}$$

$$(2) \begin{bmatrix} 1 \\ 3 \\ 5 \end{bmatrix}, \begin{bmatrix} 6 \\ 7 \\ 8 \end{bmatrix}, \begin{bmatrix} 2 \\ 0 \\ 6 \end{bmatrix}, \begin{bmatrix} 1 \\ 4 \\ 9 \end{bmatrix}$$

$$(3) \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 4 \\ 3 \end{bmatrix}$$

$$(4) \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}, \begin{bmatrix} 2 \\ 3 \\ 0 \end{bmatrix}, \begin{bmatrix} 5 \\ 8 \\ 3 \end{bmatrix}$$

Lesson 7

Part I: Transformations

Definition: A **transformation** (**function** or **mapping**)

$$T : \mathbb{R}^n \rightarrow \mathbb{R}^m$$

is a rule that assigns to each vector $\mathbf{x}$ in $\mathbb{R}^n$ a vector $T(\mathbf{x})$ in $\mathbb{R}^m$.

- $\mathbb{R}^n$ is the **domain** of T and $\mathbb{R}^m$ is the **codomain** of T .
- The vector $T(\mathbf{x})$ in $\mathbb{R}^m$ is the **image** of $\mathbf{x}$.
- The set of all images $\{ T(\mathbf{x}) \mid \mathbf{x} \in \mathbb{R}^n \}$ is the **range** of T .

Definition: A transformation T is a **matrix transformation** if, for each $\mathbf{x} \in \mathbb{R}^n$, we can write $T(\mathbf{x}) = A\mathbf{x}$, where A is an $m \times n$ matrix.

Note: For a matrix transformation $T(\mathbf{x}) = A\mathbf{x}$, the range of T is the set of all linear combinations of the columns of A .

Example 1: Let $T : \mathbb{R}^2 \rightarrow \mathbb{R}^3$ be the transformation defined by $T(\mathbf{x}) =$

$A\mathbf{x}$, where $A = \begin{bmatrix} 1 & -2 \\ 3 & 4 \\ 1 & -4 \end{bmatrix}$. Let $\mathbf{b} = \begin{bmatrix} -1 \\ 2 \\ -2 \end{bmatrix}$ and $\mathbf{c} = \begin{bmatrix} -2 \\ 1 \\ 2 \end{bmatrix}$.

(1) Find the image of $\mathbf{u}$ under T if $\mathbf{u} = \begin{bmatrix} 2 \\ 3 \end{bmatrix}$.

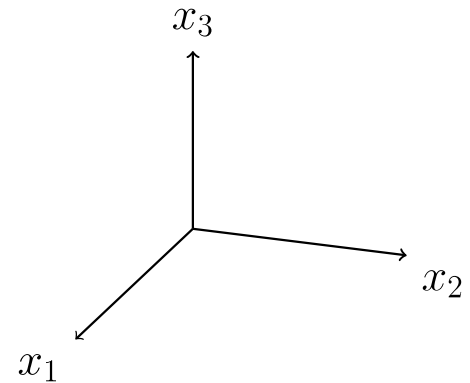
(2) Find all $\mathbf{x} \in \mathbb{R}^2$, if any, whose image under T is $\mathbf{b}$.

(3) Determine whether the vector $\mathbf{c}$ is in the range of T .

Example 2:

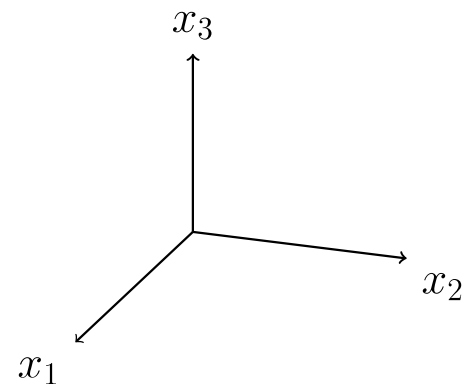
(1) Let $A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$. Show that the transformation $T : \mathbf{x} \mapsto A\mathbf{x}$

projects each point in $\mathbb{R}^3$ onto the x_1x_2 -plane. Find the domain, codomain, and the range of T .



(2) Let $A = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$. If T is the transformation so that $T(\mathbf{x}) = A\mathbf{x}$,

describe the transformation geometrically.



Part II: Linear Transformations

Definition: A transformation T is a **linear** if

1. $T(\mathbf{u} + \mathbf{v}) = T(\mathbf{u}) + T(\mathbf{v})$ for all $\mathbf{u}, \mathbf{v}$ in the domain of T , and
2. $T(c\mathbf{u}) = c T(\mathbf{u})$ for all scalar c and all $\mathbf{u}$ in the domain of T .

Note: If T is a linear transformation, then $T(\mathbf{0}) = \mathbf{0}$.

Statement: T is a linear transformation if and only if

$$T(c\mathbf{u} + d\mathbf{v}) = c T(\mathbf{u}) + d T(\mathbf{v})$$

for any scalars c, d , and for any $\mathbf{u}, \mathbf{v}$ in the domain of T .

For example, if $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ is a linear transformation that maps $\mathbf{e}_1$ to the vector $\begin{bmatrix} 1 \\ -2 \end{bmatrix}$ and maps $\mathbf{e}_2$ to the vector $\begin{bmatrix} -1 \\ 0 \end{bmatrix}$, then

$$T\left(\begin{bmatrix} 2 \\ -3 \end{bmatrix}\right) =$$

Note: A matrix transformation $T : \mathbb{R}^n \rightarrow \mathbb{R}^m$ defined by $T(\mathbf{x}) = A\mathbf{x}$, where A is an $m \times n$ matrix, is linear.

Theorem: Let $T : \mathbb{R}^n \rightarrow \mathbb{R}^m$ be a linear transformation. Then there exists a unique matrix A such that

$$T(\mathbf{x}) = A\mathbf{x} \text{ for all } \mathbf{x} \in \mathbb{R}^n,$$

and A is the $m \times n$ matrix in the form

$$A = [T(\mathbf{e}_1) \ T(\mathbf{e}_2) \ \cdots \ T(\mathbf{e}_n)],$$

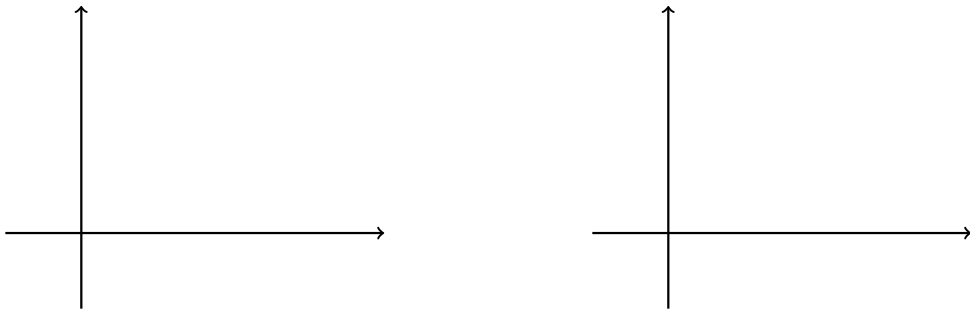
where $\mathbf{e}_1, \mathbf{e}_2, \dots, \mathbf{e}_n$ are the columns of the $n \times n$ identity matrix I .

Definition: The matrix $A = [T(\mathbf{e}_1) \ T(\mathbf{e}_2) \ \cdots \ T(\mathbf{e}_n)]$ is called the **standard matrix** of the linear transformation T .

Note: Let $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ is a transformation defined as

$$T(\mathbf{x}) = r\mathbf{x}, \quad r \in \mathbb{R}.$$

1. T is linear.
2. T is called a contraction when $0 \leq r \leq 1$ and a dilation when $r > 1$.

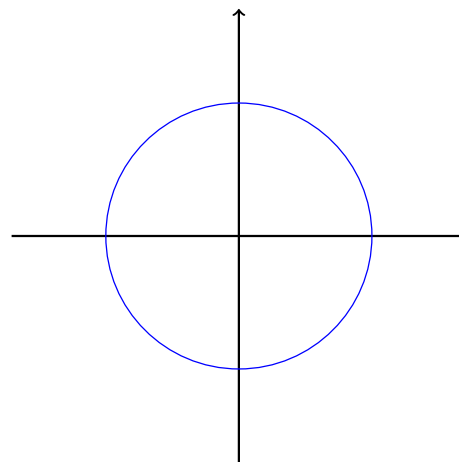


Example 3: Find the standard matrix A of the dilation transformation

$$T(\mathbf{x}) = 2\mathbf{x}, \quad \mathbf{x} \in \mathbb{R}^3.$$

Example 4: Find the standard matrix A of the linear transformation $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ that rotates each point of the plane through an angle θ .

Note: T is called a rotation transformation.



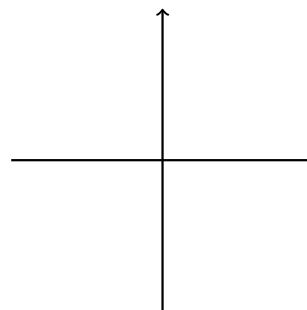
Lesson 8

Part I: Geometric Linear Transformations

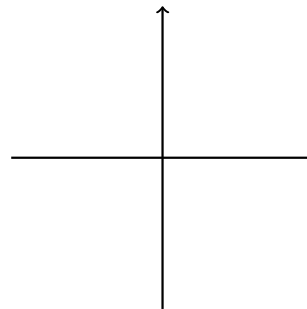
Recall: If $T : \mathbb{R}^n \rightarrow \mathbb{R}^m$ is a linear transformation, we can write $T(\mathbf{x}) = A\mathbf{x}$, where $A = [T(\mathbf{e}_1) \ T(\mathbf{e}_2) \ \cdots \ T(\mathbf{e}_n)]$ is the $m \times n$ standard matrix.

Example 1: Find the standard matrix A of each transformation T .

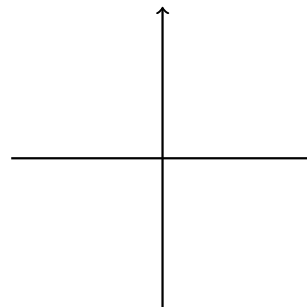
(1) reflection through the line $x_2 = x_1$:



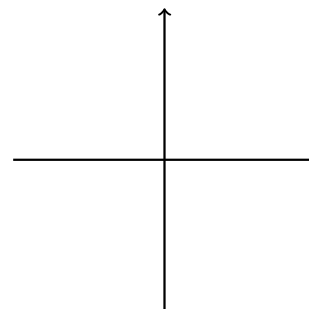
(2) reflection through the line $x_2 = -x_1$:



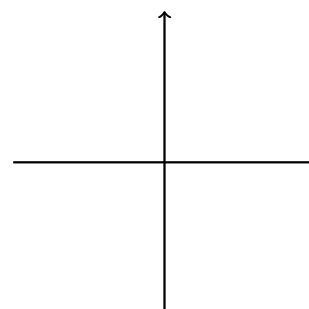
(3) reflection through the origin:



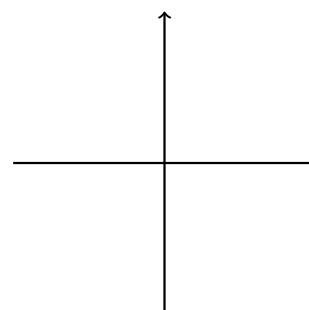
(4) reflection through the line x_2 -axis:



(5) projection onto x_1 -axis:



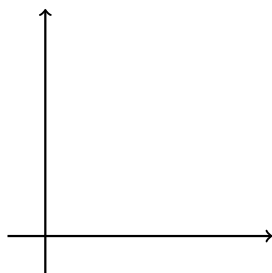
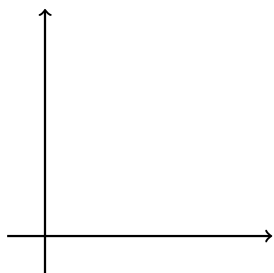
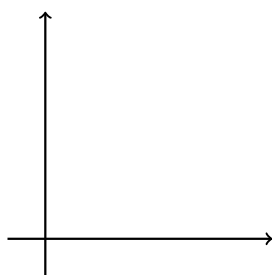
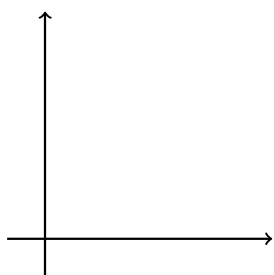
(6) vertical contraction and expansion by a factor of k ($k > 0$):



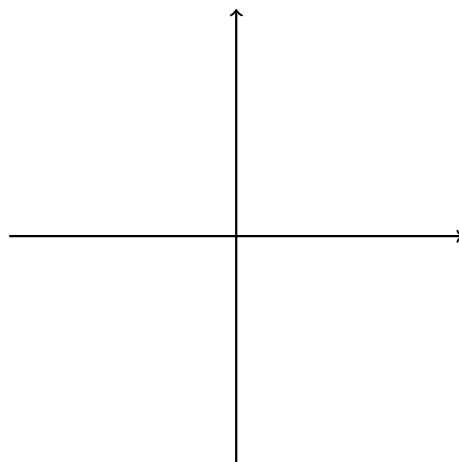
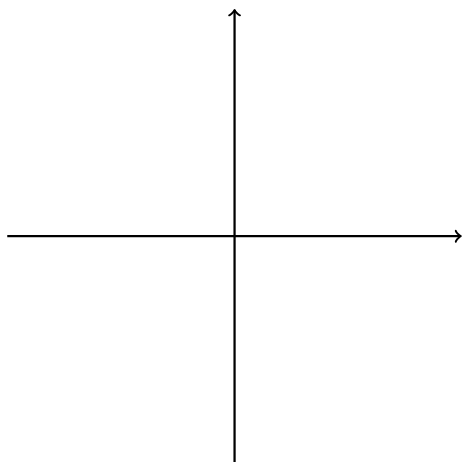
Example 2: The matrix transformations defined by the matrices

$$\begin{bmatrix} 1 & k \\ 0 & 1 \end{bmatrix} \quad \text{and} \quad \begin{bmatrix} 1 & 0 \\ k & 1 \end{bmatrix}$$

are called horizontal shear and vertical shear transformations, respectively. Find the image of the square $[0, 1] \times [0, 1]$ under each of the transformations for $k > 0$.



Example 3: Find the standard matrix A of the linear transformation $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ that first performs a horizontal shear that transforms $\mathbf{e}_2$ into $\mathbf{e}_2 + 2\mathbf{e}_1$ (leaving $\mathbf{e}_1$ unchanged) and then reflects the points through x_1 -axis.



Part II: Onto and One-to-One Transformations

Definition: A mapping $T : \mathbb{R}^n \rightarrow \mathbb{R}^m$ is a **onto** if the equation $T(\mathbf{x}) = \mathbf{b}$ has a solution for each $\mathbf{b} \in \mathbb{R}^m$.

Geometrically it means that each $\mathbf{b} \in \mathbb{R}^m$ is the image of at least one $\mathbf{x} \in \mathbb{R}^n$.

Definition: A mapping $T : \mathbb{R}^n \rightarrow \mathbb{R}^m$ is a **one-to-one** if whenever $T(\mathbf{x}) = \mathbf{b}$ has a solution for $\mathbf{b} \in \mathbb{R}^m$, the solution is unique.

Geometrically it means that each $\mathbf{b}$ in the range of T is the image of exactly one $\mathbf{x} \in \mathbb{R}^n$.

Theorem: If $T : \mathbb{R}^n \rightarrow \mathbb{R}^m$ be a linear transformation, then
 T is one-to-one $\iff T(\mathbf{x}) = \mathbf{0}$ has only the trivial solution.

Example 4: Let $T : \mathbb{R}^4 \rightarrow \mathbb{R}^3$ be a linear transformation with the
standard matrix $A = \begin{bmatrix} 1 & 3 & 2 & 0 \\ 0 & -2 & 1 & 4 \\ 0 & 0 & 0 & 3 \end{bmatrix}$.

(1) Does T map $\mathbb{R}^4$ onto $\mathbb{R}^3$?

(2) Is T one-to-one?

Theorem: Let $T : \mathbb{R}^n \rightarrow \mathbb{R}^m$ be a linear transformation and let A be the standard matrix of T . Then

1. T maps $\mathbb{R}^n$ onto $\mathbb{R}^m \iff$ the columns of A span $\mathbb{R}^m$.

2. T is one-to-one $\iff$ the columns of A are linearly independent.

Example 5: Let $T : \mathbb{R}^2 \rightarrow \mathbb{R}^3$ be a linear transformation so that

$$T \left(\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \right) = \begin{bmatrix} 2x_1 + 3x_2 \\ 4x_1 - x_2 \\ x_1 + x_2 \end{bmatrix}.$$

(1) Does T map $\mathbb{R}^2$ onto $\mathbb{R}^3$?

(2) Is T one-to-one?

Lesson 9

Part I: Matrices

Recall: We have used the following notation for an $m \times n$ matrix A :

$$A = [\mathbf{a}_1 \quad \cdots \quad \mathbf{a}_j \quad \cdots \quad \mathbf{a}_n]$$

$$\text{where } \mathbf{a}_j = \begin{bmatrix} a_{1j} \\ \vdots \\ a_{ij} \\ \vdots \\ a_{mj} \end{bmatrix} \in \mathbb{R}^m \ (j = 1 : n),$$

and we can rewrite matrix A in the form:

$$A = \begin{bmatrix} a_{11} & \cdots & a_{1j} & \cdots & a_{1n} \\ \vdots & & \vdots & & \vdots \\ a_{i1} & \cdots & a_{ij} & \cdots & a_{in} \\ \vdots & & \vdots & & \vdots \\ a_{m1} & \cdots & a_{mj} & \cdots & a_{mn} \end{bmatrix} = [a_{ij}] \ (i = 1 : m, j = 1 : n)$$

Equality of Two Matrices

Two matrices are equal if they have the same size and their corresponding entries are equal.

$$\text{i.e., } A = B \iff a_{ij} = b_{ij}, \ (i = 1 : m, j = 1 : n)$$

Sums and Scalar Multiples of Matrices

Definition: Let $A = [a_{ij}]$ and $B = [b_{ij}]$ be two $m \times n$ matrices, and let r be a scalar.

- $A + B = [a_{ij} + b_{ij}]$
- $rA = [ra_{ij}]$

Note: $A + B$ is defined only when A and B have the same size.

Properties of Sums and Scalar Multiples of Matrices

Theorem: Let A, B , and C be $m \times n$ matrices, and let r and s be scalars. Then

1. $A + B = B + A$
2. $(A + B) + C = A + (B + C)$
3. $A + 0 = A$, where 0 is the $m \times n$ zero matrix.
4. $r(A + B) = rA + rB$
5. $(r + s)A = rA + sA$
6. $r(sA) = (rs)A$

Example 1: Let $A = \begin{bmatrix} 1 & -2 & 3 \\ -2 & 4 & 1 \end{bmatrix}$, $B = \begin{bmatrix} 2 & 1 & -3 \\ 5 & 4 & -2 \end{bmatrix}$, and let 0 be the 2×3 zero matrix. Find:

$$2(A - B) + 0 =$$

Special Matrices

Definition: The **main diagonal** of a matrix $A = [a_{ij}]$ is the list of entries a_{ii} .

- A matrix $D_{n \times n}$ is a **diagonal matrix** if its non-diagonal entries are zero. (i.e., $d_{ij} = 0$ for all $i \neq j$)

Note: The $n \times n$ identity matrix I is a diagonal matrix.

- A matrix $L_{n \times n}$ is an **lower triangular matrix** if its non-zero entries are only in the lower triangle of the matrix. (i.e., $l_{ij} = 0$ for all $i < j$)

- A matrix $U_{n \times n}$ is an **upper triangular matrix** if its non-zero entries are only in the upper triangle of the matrix. (i.e., $u_{ij} = 0$ for all $i > j$)

Part II: Matrix Multiplication

Recall: If $T : \mathbb{R}^n \rightarrow \mathbb{R}^m$ is a linear transformation, we can write $T(\mathbf{x}) = A\mathbf{x}$, where A is an $m \times n$ matrix.

Statement: Let A be an $m \times n$ matrix, B is an $n \times p$ matrix, and $\mathbf{x} \in \mathbb{R}^p$. The **composition of linear transformations**

$$\begin{array}{ccccc} T : \mathbb{R}^p & \longrightarrow & \mathbb{R}^n & \longrightarrow & \mathbb{R}^m \\ \mathbf{x} & \longmapsto & B\mathbf{x} & \longmapsto & A(B\mathbf{x}) \end{array}$$

is a linear transformation

$$\begin{array}{ccc} T : \mathbb{R}^p & \longrightarrow & \mathbb{R}^m \\ \mathbf{x} & \longmapsto & (AB)\mathbf{x} \end{array}$$

If $B = [\mathbf{b}_1 \quad \cdots \quad \mathbf{b}_j \quad \cdots \quad \mathbf{b}_p]$ and $\mathbf{x} = \begin{bmatrix} x_1 \\ \vdots \\ x_p \end{bmatrix}$, then

$$B\mathbf{x} \stackrel{\text{def}}{=}$$

$$A(B\mathbf{x}) =$$

How to Calculate the Matrix Multiplication?

1. Use the definition:

Definition: If A is an $m \times n$ matrix, and if $B = [\mathbf{b}_1 \quad \cdots \quad \mathbf{b}_j \quad \cdots \quad \mathbf{b}_p]$ is an $n \times p$ matrix, then the **product** AB is the $m \times p$ matrix with columns $A\mathbf{b}_j$, $j = 1 : p$.

$$\begin{aligned} AB &= A[\mathbf{b}_1 \quad \cdots \quad \mathbf{b}_j \quad \cdots \quad \mathbf{b}_p] \\ &\stackrel{\text{def}}{=} [A\mathbf{b}_1 \quad \cdots \quad A\mathbf{b}_j \quad \cdots \quad A\mathbf{b}_p] \end{aligned}$$

Note: For matrix multiplication, the number of columns in the first matrix must be equal to the number of rows in the second matrix.

Example 2: Compute AB if the product is defined.

$$A = \begin{bmatrix} 1 & -2 & 3 \\ -2 & 4 & 1 \end{bmatrix} \text{ and } B = \begin{bmatrix} 2 & 1 \\ 1 & 4 \\ -1 & 2 \end{bmatrix}$$

2. **Row-Column Rule** for Computing AB

If A is an $m \times n$ matrix and B is an $n \times p$ matrix, the (i, j) -entry of AB is the dot product of the i th row of A and the j th column of B .

Example 3: Let $A = \begin{bmatrix} 2 & 0 & 1 \\ -1 & 1 & 3 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 2 & -3 & 1 \\ 0 & -2 & 1 & 4 \\ 1 & 3 & -1 & 1 \end{bmatrix}$

(1) Find the second column of AB .

(2) $AB =$

(3) $BA =$

Properties of Matrix Multiplication

Let A be an $m \times n$ matrix, and let B and C have sizes for which the indicated sums and products are defined. Then:

1. $A(BC) = (AB)C$
2. $A(B + C) = AB + AC$
3. $(A + B)C = AC + BC$
4. $r(AB) = (rA)B = A(rB)$ for any scalar r
5. $I_m A = A = A I_n$, where I_m and I_n are the $m \times m$ and $n \times n$ identity matrices, respectively.

Warnings:

1. In general, $AB \neq BA$.
2. The **cancellation law** does not hold for matrix multiplication. That is, if $AB = AC$, then it is not true in general that $B = C$.
3. The **zero-product rule** does not hold for matrix multiplication. That is, if $AB = 0$ (the zero matrix), we cannot conclude in general that either $A = 0$ or $B = 0$.

Unit 1 Review

Let $T : \mathbb{R}^n \rightarrow \mathbb{R}^m$ be a linear transformation so that $T(\mathbf{x}) = A\mathbf{x}$, where $A_{m \times n} = [\mathbf{a}_1 \ \mathbf{a}_2 \ \cdots \ \mathbf{a}_n]$.

- A has a pivot position in every column.

- A has a pivot position in every row.

1. Let $A = [\mathbf{a}_1 \ \mathbf{a}_2 \ \mathbf{a}_3] = \begin{bmatrix} 1 & 0 & 5 \\ -2 & 1 & -6 \\ 0 & 2 & 8 \end{bmatrix}$ and $\mathbf{b} = \begin{bmatrix} 2 \\ -1 \\ 6 \end{bmatrix}$.

(1) Determine if $\mathbf{b}$ is a linear combination of the columns of A .

(2) If yes, represent vector $\mathbf{b}$ as a linear combination of the columns of A by setting the parameter $x_3 = 0$. Is this the only representation possible?

(3) Do the columns of A span $\mathbb{R}^3$?

2. Let $A = \begin{bmatrix} 1 & -3 & 2 \\ -2 & 5 & -1 \\ 3 & -6 & -3 \end{bmatrix}$ and $\mathbf{b} = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix}$.

(1) Determine if the equation $A\mathbf{x} = \mathbf{b}$ is consistent for all possible b_1, b_2, b_3 .

(2) If not, give a description of the set of all $\mathbf{b}$ for which the equation is consistent.

(3) Describe the vectors $\mathbf{b}$ geometrically as a span of the least possible number of vectors.

3. Consider a homogeneous matrix equation $A\mathbf{x} = \mathbf{0}$, where

$$A = \begin{bmatrix} 1 & 0 & 3 & 1 \\ 0 & 2 & 1 & 4 \\ 0 & 0 & 1 & 2 \end{bmatrix}.$$

(1) Find the solutions in the parametric vector form, and describe the solutions geometrically.

(2) Does the nonhomogeneous system $A\mathbf{x} = \mathbf{b}$ has a solution for any $\mathbf{b} \in \mathbb{R}^3$?

(3) Find the general solution to $A\mathbf{x} = \mathbf{b}$ for $\mathbf{b} = \begin{bmatrix} -1 \\ 0 \\ -2 \end{bmatrix}$, and describe the solution geometrically.

4. Let $A = [\mathbf{a}_1 \ \mathbf{a}_2 \ \mathbf{a}_3] = \begin{bmatrix} 1 & -3 & 2 \\ -2 & 7 & 1 \\ -4 & 6 & h \end{bmatrix}$, and let $\mathbf{b} = \begin{bmatrix} 1 \\ -4 \\ k \end{bmatrix}$.

(1) Find h so that the set of the columns of A is linearly dependent.

(2) Find h and k so that the system $A\mathbf{x} = \mathbf{b}$ has

a unique solution:

infinitely many solutions:

no solution:

5. Let $A = [\mathbf{a}_1 \ \mathbf{a}_2 \ \mathbf{a}_3] = \begin{bmatrix} 1 & -3 & 2 \\ 3 & -8 & 8 \\ 0 & 1 & 2 \\ 1 & 0 & 8 \end{bmatrix}$, $\mathbf{b}_1 = \begin{bmatrix} 1 \\ 6 \\ 4 \\ 10 \end{bmatrix}$, and $\mathbf{b}_2 = \begin{bmatrix} 1 \\ 6 \\ 3 \\ 10 \end{bmatrix}$.

(1) Find all vectors, if any, whose images under the transformation T are vector $\mathbf{b}_1$ and $\mathbf{b}_2$, respectively, where $T(\mathbf{x}) = A\mathbf{x}$.

(2) Whether the vectors $\mathbf{b}_1$ and $\mathbf{b}_2$ are in the range of T .

(3) Does range of $T = \mathbb{R}^4$?

(4) Is there only one vector $\mathbf{x} \in \mathbb{R}^3$, for every $\mathbf{b}$ in the range of T , that maps into $\mathbf{b}$?

6. Let $T : \mathbb{R}^3 \rightarrow \mathbb{R}^2$ be the linear transformation so that

$$T \left(\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} \right) = \begin{bmatrix} x_1 - 5x_2 + 4x_3 \\ x_2 - 6x_3 \end{bmatrix}.$$

(1) Find the standard matrix A of the transformation.

(2) Determine whether the equation $T(\mathbf{x}) = \mathbf{b}$ has a solution for any $\mathbf{b}$ in $\mathbb{R}^2$.

(3) Determine whether the equation $T(\mathbf{x}) = \mathbf{b}$ has a unique solution for any $\mathbf{b}$ in the range of T .

7. True or False:

If a system $A\mathbf{x} = \mathbf{b}$ has no free variable, then it has a unique solution.